

INTRODUCTION

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WORKSPACE

relayBrain

AGENT INTEGRATIONS

5

Claude Code

your agent - cc...

Connected

Cursor

your agent - 7d...

Connected

Codex

your agent - 7e...

Connected

Copilot

your agent - 4f...

Connected

Antigravity

your agent - 2a...

Connected

RECENT AGENT ACTIVITY

8

Antigravity

The following changes w

multi_replace_file_content tool to:

c:\Users\Prakrititz Borah\Documents\GitHub\relayBrain

If relevant, proactively run terminal c

execute this code for the USER. Don't

Antigravity

The following changes w

multi_replace_file_content tool to: c:

Borah\Documents\GitHub\relayBrain

If relevant, proactively run terminal c

execute this code for the USER. Don't

Antigravity

The following changes w

multi_replace_file_content tool to: c:

Borah\Documents\GitHub\relayBrain

If relevant, proactively run terminal c

execute this code for the USER. Don't

Activity

Agent Chat

Code Edits

Coordinator

Team

Memory

COORDINATOR

Antigravity is editing app.js right now.

Editing

Readonly

Also working here

Clash

Motion

Recent editors

fs.watch

Editing now

@this machine

npm install -g relay-it

Claude Code

Idle

Cursor

Idle

Codex

Idle

Copilot

index.html

Editing now

tsconfig.json

mission-control/tsconfig.json

fs.watch has it - 44s left

app.js

Antigravity has it - 20s left

Board

List

PEOPLE

Unnath Chittimalla

gathik-jindal

Prakrititz Borah

melohub-xbit

LATEST

Claude Code — Hi

The following changes were made by

the multi_replace_file_content tool to:

c:\Users\Prakrititz...

MEMORY

84 events · 4 chats · 22 edits





You already outran Git.

12

Agents in one repo

4

Developers

0

Aware of each other

You find out at the merge.

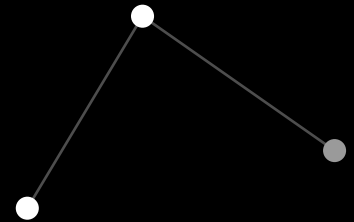
Or you don't find out at all.

Agents write at machine speed, in parallel, in the same directory, right now. They are completely blind to each other's actions.

Git assumes people. We're coordinating machines.

Git assumes commits minutes apart, one branch each. **Agents write at machine speed**, in parallel, in the same directory, right now. **There's no coordination layer built for that** until you add one.

This is **relay.it**. GitHub Desktop for AI agents.



INTRODUCTION

This isn't a hypothetical problem.

Independent studies of real agent PRs on GitHub. not a survey, not a guess.

79.4% of agent PRs overlap another agent PR in time

27.7% of all agent PRs hit a merge conflict vs. 10–20% for humans

WHEN TWO AGENT PRs OVERLAP, HOW OFTEN DO THEY CONFLICT?

Cases	Conflict rate	95% CI	Sample
Same agent, overlapping PRs	19.8%	16.8–23.2%	119 / 601
Two different agents (Cursor vs. Claude, etc.)	41.7%	33.1–50.9%	48 / 115

WHAT THEY FIGHT OVER

- 84.4% conflicted files are source code
- 57.6% both edited the same lines
- 26.8% one edited what the other deleted

Sources: [arXiv:2607.04697](#) (33,596 agent PRs, 2,807 repos) · [AgenticFlikt, arXiv:2604.03551v2](#) (142,652 agent PRs)



INTRODUCTION

What one collision actually costs.

Rate above × official vendor pricing. Not a measured average, labeled as an estimate.

PER 100 OVERLAPPING AGENT PRs

	Same agent	Different agents
Conflicts	20	42
Token \$ if redo = 1 light session (\$0.55)	\$11	\$23
Token \$ if redo = 1 heavy run (\$5)	\$99	\$209

WHERE THE MONEY ACTUALLY IS

Token cost alone is small — tens of dollars. The real spend is the parallel-agent plan:

Claude Code, active dev **\$150–250 / month**

OpenAI Codex **\$100–200 / month**

Antigravity, heavy run **up to ~\$5 / session**

You already pay **\$80–180/month** more to run agents in parallel. **20–42%** of the overlapping jobs on that plan conflict.

Relay is what makes the rest of that upgrade usable.

Sources: [Anthropic Claude Code costs docs](#) · [OpenAI Codex rate card](#) · [Google Antigravity pricing docs](#)



What `./relay` can do?





File Locks

Coordination is the core.

- When multiple agents try to edit the same file, Relay locks it
- If one agent is modifying a file, another agent cannot silently overwrite it
- Destructive conflicts are prevented before they happen and not discovered after

```
$ agent-a edit src/auth.ts  
→ lock acquired: write · auth.ts  
  
$ agent-b edit src/auth.ts  
X blocked  
  
"auth.ts is locked by Agent A  
  (Cursor, pid 4821). Waiting."
```

HOW DOES IT WORK



Dependency Graph

Beyond simple file locks.

- Files don't exist independently — a change in one can affect dozens of others
- Not just “is this file being edited?” but “does this change affect a file someone else is working on?”
- Agents keep working in parallel, without conflicts across connected parts of the project

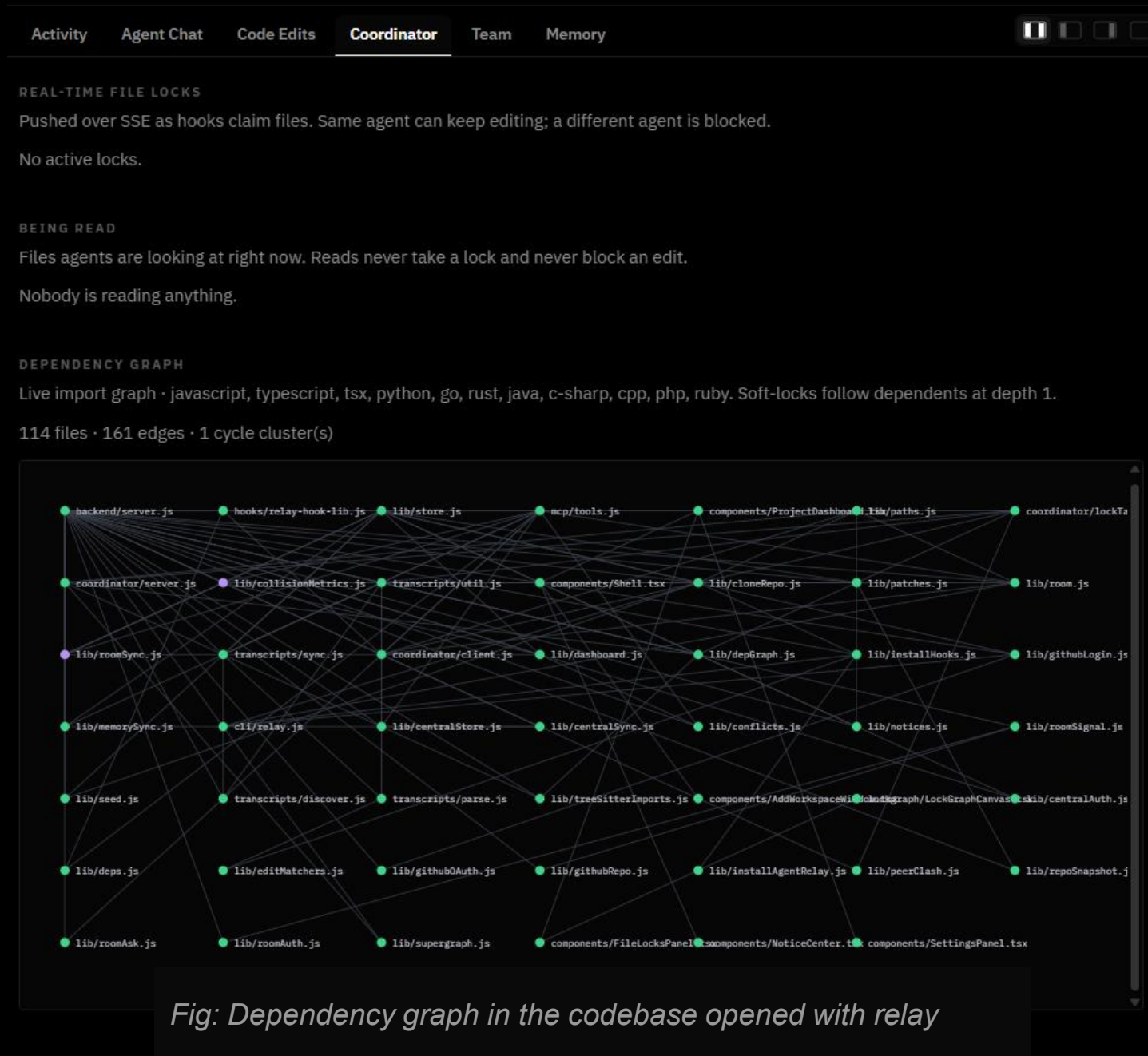


Fig: Dependency graph in the codebase opened with relay

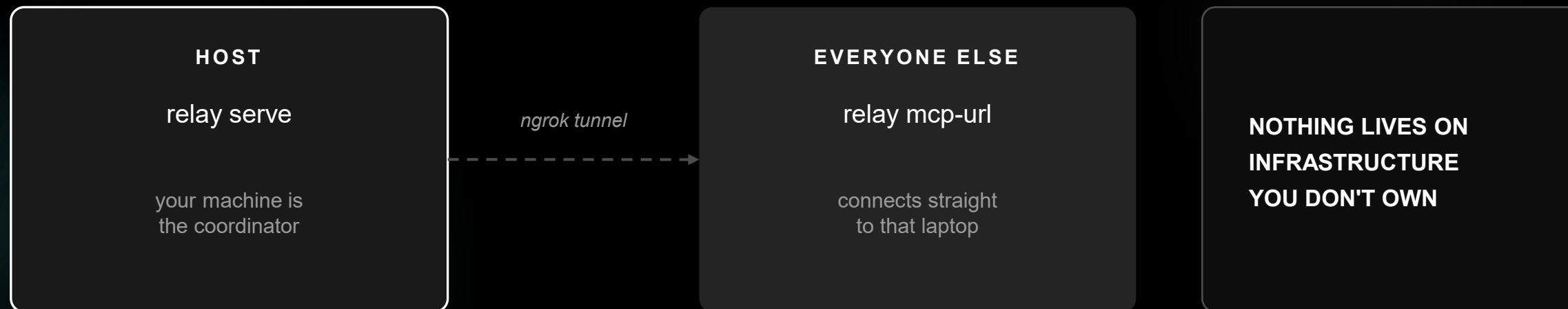


HOW DOES IT WORK



No hosted backend.

One teammate's laptop is the whole server.



You are the server. You are the client.

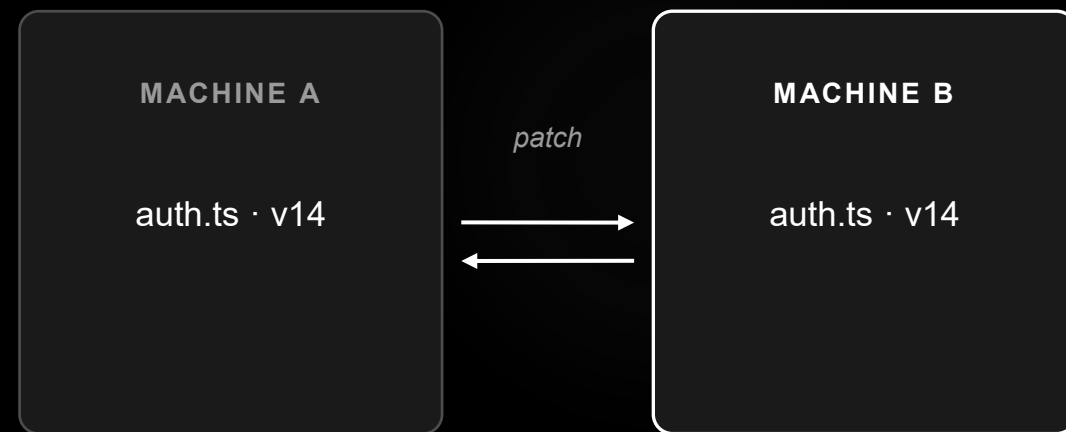


HOW DOES IT WORK

Patch Sync

Everyone stays on the same page.

- Agents aren't always on the same machine - different filesystems, environments, agents
- Relay propagates changes as patches instead of copying the whole project
- Every agent stays on the latest version of the codebase, always



same version, no full re-copy



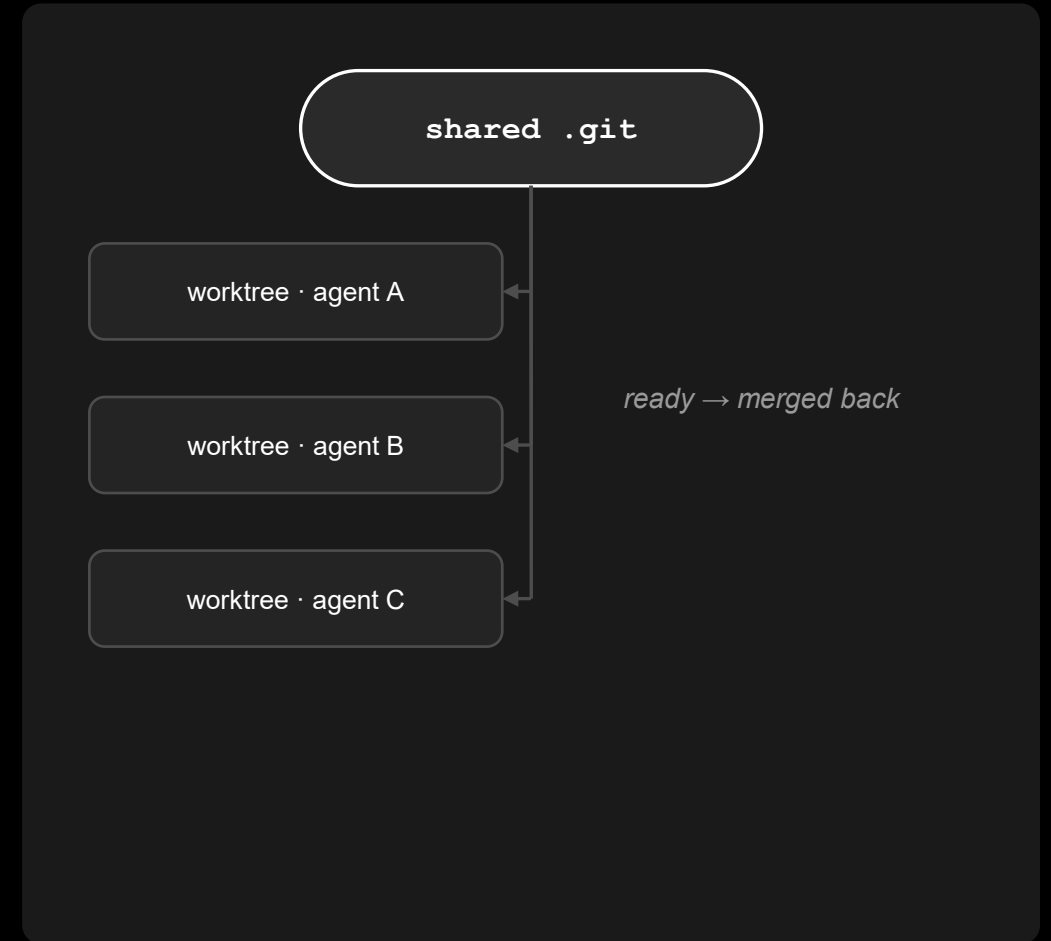
HOW DOES IT WORK



Git Worktrees

Isolated, but never disconnected.

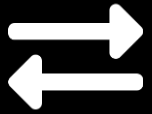
- Each agent operates in its own isolated workspace, on the same repository
- Agents experiment, modify, and test independently, without stepping on each other's changes
- When work is ready, changes integrate safely back into the main codebase



**./relay what it was vs
what it is**



COMPARISON



The Shift From Round 1

From markdown we committed to Git – to a system that stops the collision.

Before

- **A markdown update scheme** – agents wrote `.md` files; you hope everyone git pulled.
- **Git was the sync protocol** – context landed in minutes (or hours) later.
- **No locks** – two agents could trash the same file and find out at merge.
- **Single machine, solo ritual** – no room, no peers, no shared board.
- **A prototype, to prove "shared context"**.

Shipped Today

- **A coordination layer** – locks, dependency graphs, patches, live room.
- **Pre-write enforcement** – hooks claim the file, before the edit lands.
- **Patch sync across machines** – one room, same tree, no "pull and pray"
- **Live status the moment it happens** – locks, chat, collisions on one board.
- **A real product, dogfooded** - we used Relay to build Relay.

We didn't just design a coordination layer. We needed one, and it's the one we run.



AUDIENCE

Who `./relay` is built for.



Built To Move Fast

Not enterprise process — small teams shipping at agent speed.

- Startups and small teams running 2-3 agents in parallel just to keep pace
- Hackathon teams - hours on the clock, no time for a merge conflict to eat twenty minutes
- Teams already paying \$80-250/month per person to run agents in parallel, and need that spend to work



\$ who's this for

→ startups shipping multiple agents daily

→ hackathon teams racing the clock

→ teams already paying for parallel agents

if that's you, keep going.



Cut The Cost

Every conflict is a redo. Every redo is tokens and time.

- Cuts token costs - a blocked write costs nothing, a silent overwrite costs a full re-run to fix
- Keeps every collaborator's agents in sync - same lock table, same memory, same live status
- Never loses context - decisions and edits persist past a single session, for the next agent that joins



```
$ relay doctor
```

```
ok      room healthy · 3 peers
```

```
ok      0 conflicts today
```

```
move at agent speed.
```

```
don't pay for the same mistake twice.
```


HOW DOES IT WORK

Fun Facts

Initially the problem we were solving was the problem we were experiencing.



FUN FACT

We used Relay to build Relay.

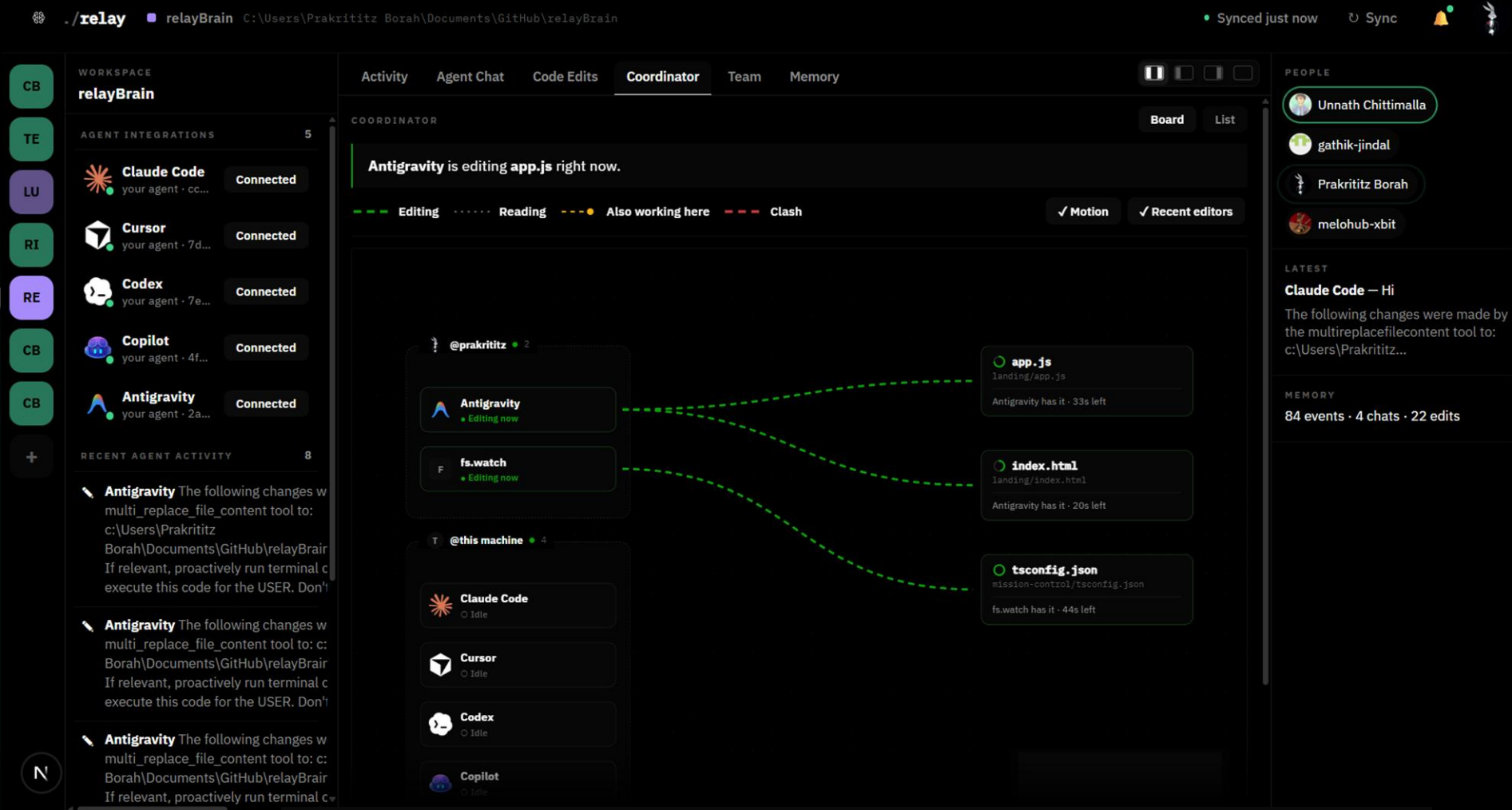


Fig: Screenshot taken during hackathon. Using relay to build relay

THANK YOU

The next time you build a hackathon project,
don't just add more AI agents.

Just./relay.it

